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SCHOOL OF SCIENCE AND APPLIED TECHNOLOGY

DEPARTMENT OF COMPUTING AND INFORMATICS

BACHELOR OF SCIENCE IN COMPUTER SCIENCE

COMP 413/424 Computer Systems Project

COURSE TITLE: COMPUTER SYSTEMS PROJECT
COURSE CODE: COMP 413
CREDIT UNITS: 3.0
PREREQUISITE: COMP 111, COMP 123, COMP 124, COMP 210, COMP 213, COMP 214, COMP 215, COMP 224, COMP 225, COMP 226, COMP 311, COMP 312, COMP 315, COMP 315, COMP 316, COMP 322, COMP 336

Purpose
To provide the learner with the opportunity to apply knowledge obtained to the solution of practical problems

Objectives
By the end of this course unit, the learners should be able to:

1. Identify different areas where applications software can be applied.
2. Plan, specify, design, implement and test a practical system to meet users' requirements.
3. Write software documentation.

Course Content
Learners work with the instructor in specifying, designing and implementing a "real world" project. Designed system useable by a department in the University. The unit runs for two semesters: Student develops the idea and presents a proposal in the first semester. Develop the actual project in the second semester. Student to present a complete project report at the end of the semester II.

Teaching Methodologies
Lectures, tutorials, supervised laboratory practical's and learners individual lab research.
Instructional Materials/Equipment
Computers with necessary software's, Whiteboard markers, Textbooks, Flip Charts, Projectors Course

assessment
Type of Assessment Weighting

Proposal	20%
Presentation	40%
Documentation	40%
Total	100%

APPENDIX I: WRITING GUIDELINES

1.1 Quality Requirements of Research Projects

The postgraduate level project demonstrates the student's readiness to join the community of scholars and practitioners in advancing the knowledge and practice in the real world of business. Consequently, students are expected to produce quality research projects:

i) Make contributions to the knowledge in the discipline,

ii) Address current problems of interest to the practitioners,

iii) Demonstrate a mastery of a specialization area within the program of study

iv) Reflect the integration of practice and scholarship, and

v) Are of publishable quality.

1.2 Research Project

1.2.1 The concept paper

Before writing proposal write a concept paper

The concept paper should be structured under the following headings:

i) Introduction/Background/Problem statement

  a) Rationale

ii) Aims and objectives

iii) Conceptual framework

iv) Method

  a) Proposed source of data

  b) Method of data collection/Instruments

  c) Method of analysis

v) Contribution to knowledge/originality concept papers should in the overall demonstrate mastery of the subject area and research methodology

The concept paper should not exceed 2500 words or 5 pages. While academic-style referencing is required (APA style), references are included in the final word count and should therefore be used strategically.

A concept paper is evaluated along the following four key dimensions:

i) Does the candidate present a set of overarching aims and objectives and research questions that focus and guide the study?

ii) Does the candidate show evidence of understanding the theory in the area in which she/he wish to conduct their study?

iii) Is the candidate able to fully conceptualize the problem

iv) Does the candidate demonstrate a clear understanding of the methods (including data sources, research instruments and methods of analysis) selected to respond to the aims, objectives and key research questions outlined in the study?

v) Is the study original and substantive enough for consideration at the study level of study? or does the student does the student demonstrate mastery of the subject area and research methodology?

1.2.2 The Research Proposal

After successful evaluation of a concept paper, the department expects each candidate to develop a research proposal for the intended research. The research proposal is a blueprint or a plan for an intended study. Research proposal preparation is essential in the development and pursuit of a research endeavor. The quality of the final research project often depends on the quality of the research proposal. Consequently, each student must develop a comprehensive research proposal before submission for oral presentation.

Size of Research Proposals

i) A project research proposal should not exceed 25 double spaced pages excluding appendices

ii) The research proposal should not exceed 35 double spaced pages excluding appendices.

The Format of the Proposal

Preliminary pages

The research proposal preliminary pages shall include the following:

i) Title page

ii) Declaration

iii) Dedication

iv) Acknowledgement

v) Table of contents

vi) List of tables

vii) List of figures

viii) List of Appendices

ix) Abbreviation and Acronyms

x) Abstract

The main body

The main body of the proposal shall normally include the following:

i) Chapter One: Introduction

ii) Chapter Two: Literature review

iii) Chapter Three: Methodology

iv) References

v) Appendices:

- *a)* Data collection instruments
- *b)* Research schedule (work plan)
- *c)* Research budget
- *d)* Any other relevant support documents

Note: The contents of these sections of the proposal are described under the Project report. The difference in the contents is in the style of writing the contents in the proposal are written in future tense while those in the projected report are written in past tense

1.3 Research Project report

Project writing should conform to the Computing and Informatics Departmental format as stated below:

1.3.1 General Requirements

(i) Typing of report

A Report must be type-written or printed on an A4 size, 80 gram bond paper. Computer /laser printed material must be of good quality. There must be consistency in the typeface using times new roman font and font size 12. The report shall be typed with double spacing using of clear UK English language.

(ii) Margins and pagination

There must be a 40mm (1.5 inches) margin on the left hand side to allow for binding, and a 25mm (1 inch) margin on the right hand side, top and bottom of the paper. Preliminary pages such as table of contents, list of tables and figures that precede the first page of text shall be numbered using small roman numerals. Pages of the main report shall be numbered consecutively in arabic numerals, starting with the first page of text and the number shall appear in the centre of the lower margin of the page.

7.3.2 The format

The research project report comprises of the following sections:

Preliminary pages

The research proposal preliminary pages shall include the following:

i) Title page
ii) Declaration
iii) Dedication
iv) Acknowledgement
v) Table of contents
vi) List of tables
vii) List of figures
viii) List of Appendices
ix) Abbreviation and Acronyms
x) Abstract

The main body

The main body of the project report shall normally include the following:

i) Chapter One: Introduction

ii) Chapter Two: Literature review

iii) Chapter Three: Research Methodology

iv) Chapter Four: Data Analysis Findings and Interpretation

v) Chapter Five: Discussions Summary and Conclusions

vi) References

vii) Appendices:

- *a)* Data collection instruments
- *b)* Any other relevant support documents

1.3.2.1 Preliminary pages

i) Spine

The spine shall be embossed in gold with the candidate's surname and initials, the abbreviation of the degree for which it has been submitted and the year. The writing shall be from bottom of the spine to top. Report is bound in Green for Master and in Grey for PhD. The face of hardcover shall be plain.

ii) Title Page

This should bear the title of the report in capital letters (21words max), followed below by the full name of the student. This is followed by a statement as follows "**A REPORT SUBMITTED TO THE COMPUTING AND INFORMATICS DEPARTMENTAL IN PARTIAL FULFILLMENT OF THE REQUIREMENTS FOR THE AWARD OF THE DEGREE OF (NAME OF THE DEGREE) , LAIKIPIA UNIVERSITY**" in the middle of the page. Finally, the month and the year of submission at the bottom and centered

Title is a concise statement of the main topic and should identify the variables.

i) Should reflect the contents of the document.

ii) Fully explanatory when standing alone.

iii) Abbreviations should not appear in title.

Example of Title Page

REVENUE DIVERSIFICATION AND COST CONTAINMENT STRAGETIES IN KENYAN PUBLIC UNIVERSITIES

ALEX KIBET

A RESEARCH PROPOSAL/REPORT SUBMITTED TO THE COMPUTING AND INFORMATICS DEPARTMENTAL IN PARTIAL FULFILMENT FOR THE AWARD OF THE DEGREE OF BACHELOR OF COMPUTER SCIENCE, LAIKIPIA UNIVERSITY

FEBRUARY, 2025

iii) Declaration

It should include both the candidate's and the supervisor's declaration duly signed. It should begin with the declaration by the candidate followed by the declaration by the supervisors as shown below:

Declaration by the candidate

This proposal/Project report is my original work and has not been presented for award of a degree in any other University or for any other award

…….. ……

Signature Date

Name of candidate and Reg. Number

Declaration by the supervisors

I/We confirm that the work reported in this proposal/thesis was carried out by the candidate under our supervision and has been submitted with our approval as university supervisors

…… ……

Signature Date

Name of supervisor, department, school and university

…… ……

Signature Date

Name of supervisor, department, school and university

iv) Dedication (optional)

A dedication statement not exceeding 25 words may follow the declaration, beginning on a separate page.

v) Acknowledgment

Acknowledgment not exceeding 150 words should follow dedication beginning on a separate page. You should acknowledge supervisors and anyone who made a significant contribution to your research technically (including materials, supplies), intellectually (assistance, advice) and financially (research funding, travel grants etc)

vi) Table of ***Contents***

A table of contents should follow acknowledgement, beginning on a new page. It is a synopsis of the structural pattern of the thesis. It contains all major sections i.e. chapter level headings. If the thesis is long, subheadings may be included. The wording and presentation (i.e. capitalization, use of special fonts and characters etc.) for all entries in the table of contents must match exactly with the text. All entries must have a corresponding page number with leader dots or dashes connecting the entry to the page number.

vii) List of Tables

A list of tables should follow the table of contents, beginning on a separate page specifying the title and the page where tables can be found in the thesis. All entries must contain a corresponding page number with lead dots or dashes connecting the entry to the page number.

viii) List of Figures

A list of figures should be on a separate page immediately after the list of tables. This should capture the title and page number of the figure where figure is found. The numbering style should be two numbers level. Titles of figures should be placed below the figure

Define terms in the text that are not common but as used in the study. Note that only uncommon terms key to the study are defined.

ix) List of abbreviations and acronyms(optional)

All symbols, abbreviations and technical terms in the thesis should be listed on this page and their full interpretations, and the units where appropriate given.

x) Abstract

The purpose of the abstract is to provide a clear and concise summary thesis. It is a brief statement of the problem, objectives of the study, target population, sampling technique, sample size, instruments, data collection, data processing and analysis, key findings and major recommendations. The abstract is required for all research projects. The abstract should have a maximum of 500 words. It should be prepared after the three chapters of the proposal or the six chapters of the project/thesis. The abstract is usually a one block (single paragraph), single spaced and without citations.

1.3.2.2 Main body of the research report/thesis

i) ***Chapter One: Introduction***

This chapter should have the following sections:

Background of the study

In this section, the researcher describes the context of the study by providing a brief discussion of the genesis of the problem, starting with the global perspective followed by the local scenario. Trends related to the problem, unresolved issues and social concerns are discussed.

Statement of the problem

The problem statement describes the need for the research study in terms of the knowledge gap to be filled. The researcher should present a clear and precise statement that indicates the gap that previous research studies have not addressed. The length of the problem statement should not exceed 1 page

Objectives of the study

The general objective- This is the purpose/goal statement which should emphasize practical outcomes of the study and should be directly linked to the title of the study.

The specific objectives – these have to be in line with the variables the candidate hypothesizes to influence the phenomena being investigated. It should be a breakdown of the general objective.

Hypothesis/Research Questions

They should be in line with the specific objectives. Research questions should be stated in the form of questions and not statements. Hypotheses when used should be stated in the null.

Significance of the study

The significance of the study is concerned with the relevance of the problem both to the practice and theory. That is, does who benefits from the study findings and how

Scope of the study

This is the kind of a caveat. It should indicate the focus of the study, geographical area or target group/population/depth of focus in concepts or variables enable the reader to make judgment about the generalizability of the findings and the time context.

Limitation of the study

This describes the caveat on generalizability of the study findings. The limitations should **NOT** be stated in terms of time or financial resources constraints.

Definition of key terms

These sections define the key terms as used in the study. The intention of the section is not to have definitions of terms whose meanings are already known. Define only the key terms used in the study whose meaning departs from the conventional usage.

ii) Chapter Two: Literature review

Literature review is a critical look at the existing research that is significant to your project. This chapter should include the following sections:

Introduction

This is an overview of contents of the chapter which should be very brief and precise.

Empirical Review (Should be organized around the study objectives/variables)

The literature review section should therefore be organized or categorized according to the research questions or specific objectives in order to ensure relevance to the research problem. You should *evaluate* what has already been done, show the *relationships* between different

works, and show how it relates to *your* project. The style of reviewing should address the following:

a) Refer to work by recognized experts in your chosen area

b) Consider and discuss work that supports and work that opposes your ideas

c) Make reasoned judgments regarding the value of others' work to your research

d) Support your arguments with valid evidence in a logical manner

e) Distinguish clearly between facts and opinions

f) Ensure the review is done chronologically

g) The hypothesized variables should be subheadings of the literature review to form a framework that would help in analysis.

h) It is not supposed to be just a summary of other people's work but critique of the existing literature relevant of the study. Review and critique any previous studies. For the review to be critical, you will need to develop critical judgment.

i) Indicate what has been done by other researchers including the methodologies used and identify the emerging Research gaps.

Theoretical review (Should be organized around the key issues identified in the preceding section)

This section should discuss current/relevant theory that supports the research problem being studied. Every variable should be anchored on the relevant theory(ies). For example in a finance problem related to capital structure, the underlying theory relevant to your project may include the following: "*The Modigliani and Miller theorem, The trade-off theory, Static trade off theory, The pecking order theory, The market timing theory, Net Income Approach"*

Note: It is not expected that the subheadings "Empirical and Theoretical review" will appear in this section. Rather the subheading will be guided by the study objectives.

Conceptual framework

This is a visual/diagrammatic representation of the relationship between the variables under study. It is informed by the empirical and theoretical review and should demonstrate an understanding of what variable influences what.

This section should begin with a brief introduction and explanation of the relationships depicted in the framework. The introduction is then followed by the presentation of the

framework. An explanation of how the framework will be used in the study should then follow.

Summary

Begin with a prelude of the facts established which serve as the basis for research gaps to be pursued. Ensure connection/flow to the next chapter.

iii) Chapter Three: Methodology

The section should provide description of the methods and procedures to be used in conducting the study. Conventionally, this chapter has the following sections:

Introduction

It is important to start this chapter with an overview of contents of the chapter. This should be very brief and precise.

Research philosophy

Describe the research philosophy upon which the study is anchored.

Research design

In this section, the researcher should identify and provide justification for the specific research design(s) or strategy(ies) used in carrying out the study. The section should emphasize on management and control of variance, justification of strategy(ies) selected, their merits and demerits. The researcher should include the procedures used to examine the variables and steps taken to control for extraneous influences that might threaten the findings of the study.

Study Area

Describe the study area and the justifications for its choice.

Target population

The researcher should identify and describe the characteristics of the population involved in the study. Population refers to the entire group of people, events, or things of interest that the

researcher wishes to investigate. Target population forms a basis from which the sample or subjects for the study is drawn. Clearly identify the population, its size and characteristics. Justify the choice of the target population. Where necessary provide the Sampling frame in the form of a table.

Sampling design

In this section, detailed description of the sample size determination and sampling method should be provided. Conventionally probability sampling techniques are used to ensure fair representation and generalization of finding to the general population. Typical examples of probability sampling include simple random sampling, systematic sampling, stratified random sampling and cluster sampling.

Data collection

The researcher is supposed to discuss the type and source of data which will be collected. This should be followed by the instruments to be used to collect the data.

The researcher should justify the chosen methods for collecting data. A description of the instruments should be given, whether they are developed by the researcher or standardized instruments. It is worth noting that for primary data the researcher can use more than one instrument, in such a case s/he should justify why more than one method is used. The section should also discuss how the ***validity*** and ***reliability*** of the data collection instruments have been addressed.

Once the data collection instruments have been developed and their validity and reliability assured, the researcher should discuss the step-by-step method of administering the data collection instruments. Issues related to research permits; research assistants should also be discussed. Finally, the research should state the approximate time and the period (dates) of collecting the data.

Data analysis and presentation

This section addresses the measurement of variables, analysis of data, hypothesis testing and/or modeling.

Measurement of variables

The criteria of measurement to be used and the hypothesized direction should be given. It should sequenced as per the objectives and summarized in a table format as shown below:

Objective	*Variable*	*Type*	*Operationalization*	*Operational definition of variable*	*Measurement*	*Hypothesized direction*
1	Market value of equity	Dependent variable	Market value of equity is taken as Market Price per Share (MPS)	MPS is the turnover per counter divided by shares traded at the end of each year	Direct measure	Increase with increased investor base
2						
3						

Data Analysis

This should present detailed steps of how the quantitative and qualitative data will be analyzed. Quantitative data should include descriptive and inferential statistics and show which of these will be used for which objective or hypothesis. Descriptive statistics include frequencies, measures of central tendencies and measures of dispersion among others. Inferential statistics involve measurement of relationships among the variables and differences between samples and thus form a basis for conclusions. Finally the section should explain how the results of data analysis will be presented and justify why the specific method of presentation should be used. The researcher is also expected to indicate the name and version of the data analysis software used.

Empirical model and hypothesis testing

It is possible to build a mathematical model solely out of the abstract concepts. This section is supposed to capture the empirical model to be used in testing the research hypothesis in studies where empirical models are used. The candidate should justify why the chosen model fits his or her study.

iv) Chapter Four:

Data analysis, presentation and interpretation

This section describes all appropriate information produced by the research procedures and statistical analyses of the finding. In presenting the results, use of tables, graphs and figures combined with a brief narrative may be more effective than using the narrative. State all significant findings in the text, referring to tables and graphs displaying all significant data. If the study has produced a large amount of raw data, do not present all of it in this section. Instead, present only the information most appropriate and relates to your research objectives and hypothesis. If appropriate, include your raw data in an appendix, referring to them within your text.

The following subheadings are advisable:

a) ***Introduction-*** summary of the structure or organization of the chapter

b) The rest of the sub sections should be guided by the objectives/research questions/hypothesis.

Note: The subsections should begin with a brief introduction, presentation of results followed by a brief interpretation.

Chapter Five: Discussion of findings, conclusions and recommendations

Introduction

Give a brief Introduction of the structure or organization of the chapter.

Discussion of findings

The discussion should be a brief essay answering the following questions

a) What are the major patterns in the observations?

b) What are the relationships, trends and generalizations among the results?

c) What are the exceptions to these trends, patterns or generalizations?

d) What are the likely causes (mechanisms) underlying these patterns resulting in predictions?

e) Is there agreement or disagreement with previous work?

f) What is the relationship of the present results to the original questions and hypotheses?

g) What are the things we now know or understand that we didn't know or understand before the present work?

h) What is the significance of the present results?

i) Revisit the theoretical/conceptual framework

The subheadings may be organized around the objectives/research questions /hypothesis of the study for logical flow.

Conclusions

With reference to the study objectives, draw conclusions, summarize the observations, interpretations and insights that have resulted from the present work. Include the broader implications of your findings and where appropriate, indicate remedial action to solve the problem.

Recommendations

Research project report/thesis should include recommendations for practice or improvement. Recommendations should be made from the conclusions.

Suggestions for further research

The researcher should provide suggestions for future research work based on the findings and conclusions generated from the study.

v) **References**

Use the American Psychological Association (APA) referencing style 7th edition at the time of publishing this document or the latest edition. APA referencing guidelines are downloadable from the internet.

Agreement of text and reference list- note that this is not a bibliography, the reference cited in the text must appear in the reference list; conversely, each entry in the reference list must be cited in the text. The author must make certain that each sourcereferenced appears in both places and that the text citation and reference list entry are identical.

vi) **Appendices:**

This should include:

a) Research Instruments

b) Figures and tables, including captions, should be embedded in the text and not in an appendix, unless they are more than 1-2 pages and are not critical to your argument.

Note: Appendices should be sequenced in the order in which they appear in the text.